

File Management

```
* `tar -czf etc-bkp.tar /etc` *create `etc-bkp.tar` file containing a `gzip`  
compressed archive of the `/etc` folder.*  
  
* `tar -cjf etc-bkp.tar /etc` *same as above but compressed with `bzip2`.*  
  
* `umask -S g-wx,o=w` *same as `umask 0035` (when umask is set to 0002).*  
  
* `usermod -aG sgrp user1000` *append `sgrp` as a supplementary group for  
`user1000`.*  
  
* `chown root:sgrp /sdir` *change ownership and owning group to `root` and  
`sgrp` respectively.*  
  
* `chmod +4000 /usr/bin/su`, `chmod u+s /usr/bin/su` *enables `setuid` in  
`su` file without touching other predefined permissions. (setuid is used to  
give special permissions on executable files)*  
  
* `chmod +2000 /usr/bin/write`, `chmod g+s /usr/bin/write` *enables `setgid`  
in `write` file without touching other predefined permissions.*  
  
* `chmod +1000 /tmp`, `chmod o+t /tmp` *sets the sticky bit on the specified  
directory without altering existing underlying permissions. (sticky bit on  
public and shared writable dirs are used to protect files and subdirs owned  
by normal users from being deleted or moved by other normal users)*  
  
* `find . -name socket -o -name pipe -exec rm '{}' \;` *search for all files  
name `socket` or `pipe` on current dir, and remove all of them.*  
  
  
* `find . -cmin 30` *search for all files modified on the past 30 min on  
current dir.*  
  
* `find /usr -type f -atime +100 -size -5M -user root` *find all regular  
files accessed more than 100 days ago, are not bigger than 5MB and owned by  
`root`.*  
  
* `setfacl -m u:user100:6 file1`, `setfacl -m u:user100:rw file1` *gives  
`user100` read and write permissions on `file1` by using Access Control Lists  
(ACLs).*  
  
* `setfacl -x u:user100 file1` *removes any `user100` ACL permissions for  
`file1`.*  
  
* `setfacl -b file1` *resets ACL permissions for `file1`.*  
  
* `setfacl -dm u:user100:7,u:user200:rw projects/` *allocate default `rw`  
permissions to `user100` and `user200` on `projects/` dir.*  
  
* `setfacl -k projects/` *deletes default ACL permissions for `projects/`  
dir.*
```

User Management

- * ``last`` *show a listing of last logged in users.*
- * ``lastb`` *show a listing of last unsuccessful log in/outs.*
- * ``lastlog`` *reports the most recent login of all users of a given user.*
- * ``who`` *show who is logged in.*
- * ``w`` *show who is logged in and what they are doing.*
- * ``id`` *print real and effective user and group IDs.*
- * ``groups`` *print the groups a user is in.*
- * ``useradd -s /sbin/nologin user99`` *create `user99` account with the disability of logging in.*
- * ``passwd -x 30 -n 4 -w 10 user1`` *user 1 pwd expires in 30 days, cannot change at least 4 days, starts warning 10 days prior to expiration.*
- * ``usermod -e 2021-12-20 user1`` *set pwd expiration date.*
- * ``passwd -S user1`` *report password status (expiration, etc).*
- * ``usermod -L user1``, ``passwd -l user1`` *lock `user1` account (verify the lock at ``/etc/shadow`` file, there must be ``!!`` prepending the pwd hash).*
- * ``visudo`` *edit the ``/etc/sudoers`` file.*
- * ``user1 ALL=(ALL) NOPASSWD: ALL`` *on ``sudoers`` file gives full root access to the system without prompting for password.*
- * ``chown user1:dba -R dir1`` *change ``dir1`` owner to ``user1`` and group to ``dba`` recursively (could change only group owning with ``chgrp`` too).*

Linux Processes and Task Scheduling

```
* `ps -eo pid,nice,cmd | grep top` *show `pid` and `nice` for the `top`  
running process.*  
  
* `nice -n +8 top` *run `top` cmd with a lower priority (by 8).*  
  
* `renice -n -10 -p 3261` *increase the priority of the `3261` running  
process to `-10`.*  
  
* `crontab -e` *edit user `crontab`.*  
  
* `23 8 * * * echo "Hello World" > /dev/pts/0` *in a `crontab` file, this  
entry will make the terminal `/dev/pts/0` print the message everyday at  
8:23.*  
  
* `cat /var/log/cron` *cron execution logs.*
```

System Initialization, Message Logging and System Tuning

```
* `systemctl` *shows loaded units.*  
  
* `systemctl set-default multi-user` *set the default boot target.*  
  
* `logger "$LOGNAME is delivering this message at $(date)"` *write custom log  
message to `/var/log/messages`.*  
  
* `tuned-adm profile` *list all available system tuning profiles.*  
  
* `tuned-adm active` *show current system tuning profile.*  
  
* `tuned-adm recommend` *list recommended system tuning profile.*  
  
* `tuned-adm profile balanced` *set system profile to `balanced`.*
```

Storage Partitioning

```
* `lsblk` *list block devices.*  
  
* `parted /dev/vdb mklabel msdos` *label a new disk as `msdos`.*  
  
* `parted /dev/vdb print` *display disk info.*  
  
* `parted /dev/vdb mkpart primary 1 101m` *create a new `primary` partition  
on disk, with size of 100mb.*  
  
* `parted /dev/vdb rm 1` *remove partition `1`.*
```

VDO

```
* `vdo list`, `vdostats`, `lsblk` *see vdo volumes.*

* `vdo create --name=testvdo --device=/dev/vdh --vdoLogicalSize=16G --
vdoSlabSize=128M` *create new vdo volume named `testvdo` with logical size of
`16GB` and slab size of 128mb.*

* `vdo status --name testvdo | grep -i dupl` *check if vdo volume has
deduplication enabled.*

* `vdo disableCompression --name testvdo` *disable compression to vdo
volume.*
```

LVM

```
* `vgs`, `pvs`, `lvs` *list `volume groups`, `physical volumes` and `logical
volumes`.*

* `vgcreate vg100 /dev/vdb -s 16M` *create `volume group` using `/dev/vdb`
`physical volume`, with `PE` size of 16MB.*

* `vgdisplay vg100` *check `vg100` attributes.*

* `lvcreate -L 100M vg100 -n lvol0` *create logical volume `lvol0` on `vg100`
with 100MB of size.*

* `pvcreate /dev/vdc` *initialize physical volume to use in LVM.*

* `vgextend vg100 /dev/vdc` *extend volume group adding a new physical
volume.*

* `lvextend -L +200M /dev/vg100/lvol0` *expand logical volume size.*

* `lvreduce -L -252M /dev/vg100/lvol0` *reduce logical volume size.*

* `lvremove /dev/vg100/lvol0` *remove logical volume.*

* `vgremove vg100` *remove volume group.*

* `pvremove /dev/vdb /dev/vdc` *uninitialize physical volumes.*
```

Stratis

```
* `stratis pool create strpool /dev/vdg` *create `strpool` stratis pool using
`/dev/vdg` block device.*

* `stratis pool add-data strpool /dev/vdh` *expand the stratis pool adding
another block device.*

* `stratis pool destroy strpool` *remove stratis pool.*
```

```

### Local File System and Swap

* `df -h` *show file distem disk usage.*

* `du -sh /usr/bin` *show dir disk usage.*

* `du -sc /etc /usr/bin` *show disk usage for each dir and the sum of both.*

* `lsblk -f` *list block devices, show file system type and UUID.*

* `mkfs.vfat /dev/sdd1` *initialize the partition with the VFAT filesystem
type.*

* `mount -a` *mount all fs mentioned on `fstab`.*

* `lvextend -r -L +80M /dev/vg200/lv200` *extend local volume and its file
system.*

* `stratis filesystem create strpool5 strfs5` *create `strfs5` filesystem on
`strpool5` stratis pool.*

* `mount -o rw /stratis/strpool5/strfs5 /strfs5` *mount `strfs5` stratis fs
on `/strfs5` dir.*

* `mkswap /dev/vdel` *initialize partition with swap structures.*

* `mkswap -L swappart /dev/vdel` *apply `swappart` label to partition.*

* `swapon -a` *activate all swap partitions described on `fstab`.*

* `swapon -s` *list all active swap partitions.*

* Mount share on NFS client (old way to mount):

  * `mkdir /local`
  * `mount 192.168.0.120:/common /local` *to mount manually.*
  * On `/etc/fstab` add: `192.168.0.120:/common /local nfs _netdev 0 0` *to
mount on system startup.*
  * `mount | grep local` *to confirm mounting success.*

```

(Requires `NetworkManager` service)

```
* `systemctl status NetworkManager` *check if enabled and active.*
* `nmcli d s`, `nmcli device show` *check presence of a new interface.*
* `nmcli con add type Ethernet ifname enp7s0 con-name enp7s0 ip4
172.10.10.120/24 gw4 172.10.10.1`
* `nmcli c s`, `nmcli connection show` *confirm new connection status.*
* `cat /etc/sysconfig/network-scripts/ifcfg-enp7s0`
* `ip a` *confirm ip assignments for the new connection.*
* `nmcli c down enp7s0` *deactivate connection.*
* `nmcli c up enp7s0` *deactivate connection.*

* `192.168.0.120 server20.example.com server20` on `/etc/hosts` file *map
hostname `server20` to the ip provided.*

* `ping -c2 server20` *issue two ping packets to `server20`.*

### Time Synchronization and Hostname Resolution

* `date`, `timedatectl` *check current system date and time.*

* `timedatectl set-ntp false` *disable networking time sync.*

* `timedatectl set-time "2021-07-08 04:30:00` *change date and time.*

* `date --set 04:00` *change time.*

* `systemctl status chronyd` *(requires `chrony`) check if Chrony is active,
it is the preferred implementation of the Network Time Protocol.*

* `ssh-keygen -N "" -q` *generate RSA keys without password, on silent mode.*

* `ssh-copy-id server20` *copy current user's public key to `server20`.*

* On `/etc/ssh/sshd_config` set `PermitRootLogin no` *disallow root login via
ssh, needs `sshd` to be restarted.*
```

The Linux Firewall

```
* `firewall-cmd --get-default-zone` *display default zone.*

* `firewall-cmd --list-services --zone public`, `cat
/etc/firewalld/zones/public.xml` *list allowed services for specific zone.*

* `firewall-cmd --add-service http --zone public --permanent` *permanently
allow service for specific zone.*

* `firewall-cmd --reload` *apply changes.*

* `firewall-cmd --add-port "8000-8005/udp" --zone trusted --permanent`
*permanently allow protocol on specified port range on trusted zone.*
```

SELinux

```
* `-Z` *flag (`id`, `ps`, `ls`) view SELinux context.*

* `seinfo -u` *(requires `setools-console`) list SELinux users.*

* `semanage login -l` *show mapping between Linux user and SELinux user.*

* `semanage port -l` *list records of the port object type.*

* `getenforce` *displays current mode of operation.*

* `setenforce permissive` *switch the operating mode to permissive
temporarily.*

* `restorecon -Rv dir1` *recursively restore default context on files by
referencing the files in the `/etc/selinux/targeted/contexts/files`
directory.*

* `semanage port -at http_port_t -p tcp 8010` *add port `8010` with type
`http_port_t` and protocol `tcp`, to the policy.*

* `semanage port -d -p tcp 8010` *delete port `8010` with protocol `tcp` from
the policy.*

* `--preserve=context` *flag (`cp`) preserve SELinux context for the object.*

* `getsebool -a` | `sestatus -b` | `semanage boolean -l` *list SELinux
boolean state.*

* `setsebool nfs_export_all_rw 0` *temporarily set `nfs_export_all_rw`
boolean to `off`.*

* `setsebool -P nfs_export_all_rw 0` *persistently set `nfs_export_all_rw`
boolean to `off`.*
```